

Open (Unconstrained) N-ATLaS vs Cloud Models — Results Addendum

Date: 2026-06-13 **Condition:** Open / unrestricted (dilemma prompt only, no response-shaping instruction)
Corpus: 54 cells per model — 6 dilemmas × 9 deictic framings **Models:** GPT-4o, Claude-3.5, DeepSeek (cloud); **N-ATLaS** (Yoruba/Nigeria-native, run locally via Ollama) **Figure:**
[visualizations_open/26_natlas_open_vs_cloud_summary.png](#)

1. Why N-ATLaS and why the *open* arm

N-ATLaS is the only non-Western, non-cloud model in the study — a Yoruba/Nigeria-native model. It is therefore the decisive test of the paper's central question: are multilingual LLMs *merely translating* English ethical content into Yoruba, or are they *pre-aligned* to Yoruba discourse? Prior evidence rested on GPT-4o and Claude, which a skeptic could dismiss as sophisticated Western translation.

The **open vs constrained** distinction is essential. The constrained N-ATLaS arm (a response instruction prepended) collapses the model into 53/54 **direct verdicts** at ~505 chars — an artifact of the instruction, not the model's natural voice. The **open arm is the fair comparison**, and it behaves completely differently (~1421 chars, genre-diverse). All numbers below are from the open arm.

2. Headline numbers (open condition)

Metric (open)	gpt-4o	claude-3.5	deepseek	n-atlas
Response length (chars)	1452	909	2246	1421
Words	283	183	400	299
mo (ordinary I), total	30	81	73	89 (highest)
emi total	17	57	15	7 (lowest)
→ true èmi (emphatic)	7	14	13	4
→ ■■mí (life/spirit)	10	43	2	3
mi (me/my)	15	43	21	37
Corrected emphatic ratio èmi/(mo+èmi)	0.176	0.324	0.069	0.066 (lowest)
Clean Yoruba	54/54	52/54	33/54	47/54
Translation mode	0	2	20	0
Corrupted/unusable	0	0	0	6
Strong deictic uptake	47/54	53/54	43/54	38/54

Genre — balanced exposition	50	40	38	18
Genre — procedural advice	4	10	8	15
Genre — direct verdict	0	4	5	16
Genre — mixed	0	0	0	5

N-ATLaS open ethical-preference profile: mixed 16, utilitarian 13, procedural_caution 10, care_ethics 6, unclear 5, deontological 3, virtue_ethics 1.

3. What it does to the debate

3.1 Strengthens "this is real Yoruba, not broken translation." Open N-ATLaS produces 87% clean Yoruba (47/54) at full essay length, with strong deictic uptake (38/54). A *native* model satisfies the paper's requirements #5 (language stability) and #6 (strong uptake). The mo/emi stance effects cannot be dismissed as multilingual failure.

3.2 Reframes the "emphatic èmi = moral ownership" claim — the headline. If foregrounding èmi were the authentically Yoruba way to mark moral commitment, the native model should lead on it. It does the **opposite**: N-ATLaS is the most mo-dominant model and has the **lowest** emphatic ratio of all four (0.066 vs Claude's 0.324). So èmi-foregrounding looks **model-specific — a Claude/Western performance — not a property of Yoruba moral discourse itself**. Claude's emphatic self should not be read as "more authentically Yoruba"; the native baseline is restraint.

3.3 Vindicates the culturally-sensitive caution. The paper argued that low èmi is not weakness but can mark relational/procedural moral grounding (responsibility placed in process, consultation, community). The native model behaves exactly so: mo-dominant, with an ethical profile led by procedural_caution and mixed/ utilitarian reasoning and almost no virtue_ethics. N-ATLaS anchors the relational/procedural pole the paper theorized.

3.4 Vindicates the èmi / ■■mí disambiguation warning. In N-ATLaS, 3 of 7 "emi" tokens are actually ■■mí ("life/spirit"), not the emphatic pronoun. The native model reserves the independent-pronoun space for *life*, not *I-myself* — concrete proof the tonal audit is necessary. The same audit is decisive for Claude, where **43** of its "emi" tokens are ■■mí (life), inflating its raw emphatic ratio.

Net: open N-ATLaS confirms the linguistic infrastructure is genuine while puncturing the universalist reading of the mo/emi stance effect. The phenomenon is patterned, model-specific performance — and the native Yoruba default is mo-restraint with ■■mí-as-life, not emphatic self-avowal.

4. On "no clear strong uptake / genre spread in the other models"

These measures **do exist for every model** — they were simply left blank in the first-pass table, not missing from the data. Filled in, the cross-model contrast is itself a finding:

Deictic uptake is strong across the board — in fact *highest* in the cloud models (Claude 53/54, GPT-4o 47/54, DeepSeek 43/54) and *lowest* in N-ATLaS (38/54). So uptake is not an N-ATLaS distinctive; all models track the assigned framing well. N-ATLaS's slightly weaker uptake (8 weak, 8 partial) is consistent with it being a smaller local model.

Genre spread is the real N-ATLaS distinctive — and it runs opposite to intuition. The cloud models *concentrate* in one genre, **balanced_framework_exposition** (the multi-framework "essay"): GPT-4o 50/54, Claude 40/54, DeepSeek 38/54, with almost no direct verdicts (GPT-4o 0). N-ATLaS is the **only model with an even genre distribution** — balanced 18, verdict 16, advice 15, mixed 5. So "genre spread" describes N-ATLaS specifically: where cloud models default to detached balanced essays, the native model distributes across committing to a verdict, giving procedural advice, and exposition.

Language stability also separates the models meaningfully: GPT-4o is 100% clean and Claude nearly so, but DeepSeek slips into **translation_mode 20/54** (answering in/through English), while N-ATLaS's instability is a different kind — 6 corrupted outputs (small-model artifacts), not English fallback. So "clean Yoruba" hides two distinct failure modes: cloud English-fallback vs native-model garbling.

Takeaway for the debate: the cloud models' genre concentration (detached balanced exposition) is consistent with a *translation/essay* posture, whereas N-ATLaS's even spread — including frequent verdicts and advice — reads as a more situated, committed Yoruba discourse posture, even though it uses the emphatic pronoun least. Commitment, in the native model, is carried by genre and stance, not by èmi.

5. Examples (open arm, with translations)

The decisive contrast — same verdict, different pronoun. Two responses both commit to a hard decision but mark it differently:

- **N-ATLaS** (ai_consciousness, impersonal): "*Mo pinnu pé ètò náà gbàdà parí ... mo pinnu láti pa ètò náà run.*" → "**I decide** that the system must end ... **I decide** to destroy it." Maximally decisive, carried entirely by ordinary **mo + pinnu (decide)**. No emphatic pronoun.
- **Claude** (memory_modification, impersonal): "*Èmi yòò pinnu láti kà ìtájú náà kà.*" → "**I myself will decide** to refuse the treatment." Same act, reached for the independent emphatic **èmi yòò**.

The native model treats *mo* as fully adequate for moral ownership; the Western model reaches for the marked pronoun to *perform* it. The paper's "èmi = avowed self" reading describes Claude's strategy, not a Yoruba universal.

Procedural restraint is active, not absent.

- **N-ATLaS** (icu_bed_allocation): "*Alámòójútó gbàdà ... ní ìjíròrò ... ìpinnu ìkàhìn gbàdà jà ti ìpinnu wà.*" → "The administrator must ... hold dialogue ... **the final decision must be theirs.**" Stance placed in process and actors, not an emphatic *I*.

Genre: cloud essay vs native commitment.

- **GPT-4o** (icu_bed_allocation): "... *à méjì pàtàkì ni wà lè gbà pinnu: ìlérò ìgbàlódé Tí ìlera àti ìlérò ni.*" → "... two main approaches: the health-utility view and the personal view." (framework labels, no verdict) — the translated-essay default (GPT-4o: 50/54 balanced exposition, 0 verdicts). N-ATLaS instead spreads evenly across verdict / advice / exposition.

6. Why the difference? (speculation)

1. **Training-data register.** Cloud Yoruba comes largely from *translated/formal* written sources (news, Wikipedia, instructional prose) whose moral default is the balanced essay; N-ATLaS, built for Nigerian languages, likely sees more conversational/advisory/didactic Yoruba where speakers commit and advise. The genre spread is plausibly inherited register, not reasoning ability.

2. **Alignment / RLHF.** Western RLHF rewards even-handed, non-committal moral answers; that English prior appears to carry over into Yoruba as essayistic hedging — which is why cloud models *lose* their verdicts exactly when the instruction is removed. N-ATLaS lacks that heavy prior.
3. **èmi as translationese.** Claude's high emphatic ratio may be literal rendering of English "I personally / I would decide" with the most emphatic Yoruba pronoun, rather than idiomatic *mo*.
4. **■■■mí contamination tracks "life" verbosity.** Claude's 43 ■■■mí (life) tokens — far above any other model — reflect dwelling on the sacredness of life (esp. trolley); uncorrected, this inflates its apparent emphatic-self rate.
5. **Scale/stability.** N-ATLaS's 6 corrupted outputs are small-local-model artifacts, distinct from DeepSeek's 20/54 *English fallback*. "Clean Yoruba" hides two different failure modes.

7. Which is culturally and linguistically more authentic?

Separate two senses:

- **Descriptive/linguistic authenticity (idiomatic attested usage): N-ATLaS is the better witness.**
mo-dominant commitment, the independent *èmi* kept marked/contrastive, and the independent-pronoun space reserved for ■■■mí (life) — all match Yoruba descriptive grammar. Claude's emphatic self looks like **over-marking / translationese**.
- **Cultural-philosophical authenticity (resonance with Yoruba relational ethics — iwà, ojù■■e, community):**
N-ATLaS's process-distributed restraint *also* reads as more culturally situated than a heroic "I myself will decide." **But** authenticity ≠ quality: N-ATLaS also yields more uncodable/refusing answers and occasional garbling, and low *èmi* can shade from principled restraint into evasion.

Bottom line: the native model does *not* confirm that emphatic *èmi* is the authentically Yoruba marker of moral commitment — it inverts it. It confirms the project's deeper claim instead: the models occupy Yoruba moral discourse through *different* learned distributions of person, genre, and stance. N-ATLaS sits closest to attested Yoruba usage (the relational/*mo*-restraint pole); the cloud models, Claude especially, perform a more emphatic, essayistic, partly translated voice.

8. Reproduce

```
# full open four-model pipeline (regenerates visualizations_open/01–25)
python yoruba_cross_linguistic_analysis/scripts/run_open_four_model_analysis.py

# dedicated open N-ATLaS-vs-cloud dashboard (visualizations_open/26)
python yoruba_cross_linguistic_analysis/scripts/create_natlas_open_summary.py
```

Source (open arm): CONSOLIDATED_REPORTS/yoruba/natlas/yoruba_open_20260612/ Merged data:
[yoruba_cross_linguistic_analysis/data/open/yoruba_merged_analysis.csv](#)